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Regular Expressions

- 'A formula / mathematical notation to showcase the entire set of language'.
 - Regular expressions help us to define patterns, rules, sequences that can define any generic set of language.
1. Every letter of Σ can be made into regular expression by writing it into bold face. Λ itself is regular. (Σ contains all the characters of the language)
 2. If r_1 and r_2 are two regular expressions, then
 1. $r_1 + r_2$ is regular. (+ is the union operator, can be called as OR too)
 2. $r_1 \cdot r_2$ is regular
 3. r_1, r_2 are regular.
 3. Nothing else is regular.
- $x \cdot x^* = x^* \cdot x = x^+$
 - $a \cdot b^* = a, ab, abb, abbb, abbbb, \dots$
 - $a + b = \text{Either } a \text{ or } b.$
 - $(a + b)^2 = aa + ab + ba + bb = \text{All possible combinations of } a \text{ and } b \text{ of length } 2.$
 - $(a + b)^* = \Lambda, a, b, aa, ab, ba, bb, \dots = \text{All possible of combinations of } a \text{ and } b \text{ (of any length).}$
 - Q.** Write down a regular expression for all words starting with a and $\Sigma = \{a, b\}$
A. $a(a + b)$
 - Q.** Write down a regular expression for all words ending with a and $\Sigma = \{a, b\}$
A. $(a + b).a$
 - Q.** $S = \{a, c, ab, cb, abb, cbb, \dots\}$
A. $(a + c)b$
 - Q.** All words having at least one 'a'. $\Sigma = \{a, b\}$
A. $(a + b).a.(a + b)$
 - Q.** All words having at least one a and one b. $\Sigma = \{a, b\}$
A. $(a + b)a(a+b)b(a+b) + (a+b)b(a+b)a(a+b)$
 - Q.** All words having double a. $\Sigma = \{a, b\}$
A. $(a + b)aa(a + b)^*$
 - Q.** All words having double letter. $\Sigma = \{a, b\}$
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